

Lecture-by-Lecture Plan

L	UNIT 1: DC Circuits & Theorems	L	UNIT 2: AC Circuits & 3-Phase	L	UNIT 3: Transformers & DC Machines	L	UNIT 4: AC Machines & Power Plants
1	Ohm's Law & Kirchhoff's Laws (KCL/KVL)	1	Sinusoidal Signals: Peak, Avg & RMS	1	Transformer: Principle & Construction	1	1-Phase Induction Motors: Principles
2	Series, Parallel & Mixed Circuits	2	Form Factor & Peak Factor	2	1-Phase vs 3-Phase (Core/Shell)	2	Capacitor & Resistance Start Motors
3	Mesh Current Method of Analysis	3	Phasors: Rectangular & Polar Forms	3	EMF Equation & Ideal Transformer	3	3-Phase Induction: Squirrel Cage
4	Nodal Analysis Method	4	Trig. & Exponential Phasor Forms	4	Losses, Efficiency & Regulation	4	Slip Ring Induction Motors
5	Superposition Theorem	5	Analysis of Series RLC Circuits	5	DC Machine: Working & Construction	5	Synchronous Generators: Principles
6	Thevenin's Theorem	6	Analysis of Parallel RLC Circuits	6	EMF Equation of DC Machine	6	Generator Types & Construction
7	Norton's Theorem	7	Real, Reactive & Apparent Power	7	DC Motors: Types & Working	7	Thermal, Diesel & Gas Power Plants
8	Maximum Power Transfer Theorem	8	Power Factor & its Significance	8	Applications of DC Motors	8	Nuclear & Hydroelectric Plants
9	Star-Delta Transformation I	9	3-Phase: Star & Delta Connections	-- -	---	9	Solar, Wind & Fuel Cell Energy
10	Star-Delta Transformation II	10	Two-Wattmeter Method (3-Phase)	-- -	---	10	Tidal & Geo-thermal Power